



4 SAFETY INSTRUCTIONS

4.1 Machine Usage

RAUMAK, committed to the safety of the operators, recommends the use of the equipment with maximum attention, carefully following the procedures described in this chapter. To ensure safety, the use of PPE (Personal Protective Equipment) is also recommended.

RAUMAK machines are designed in accordance with customer specifications, respecting all safety rules. Various devices are installed in order to reduce and, if possible, eliminate the risk of accidents and harm to users.

Raumak is not responsible for accidents or damage caused by misuse by the operator, inadequate training (person not responsible for the equipment) or for not observing the safety rules and procedures described in this manual.

4.2 Work Environment

The customer is responsible for adapting his environment to the installation of the machine, i.e., provide sufficient space for the equipment, movements and maintenance, environment with low humidity, adequate lighting and level floor with capacity resistant to the weight of the equipment.

It is extremely important that the electrical and compressed air points are already properly installed (as described in the technical delivery form). The equipment should not be installed where it may be exposed to the flow of water, oils and other liquids. The machine is not designed to operate in a humid, corrosive and/or explosive environment. Sources of heat or direct sunlight on the equipment should be avoided.

4.3 Fundamental Rules for Working Safely

To avoid risky situations, it is necessary that all those responsible for the equipment read this manual with the utmost care, being sure that they understand it. A daily inspection of the safety functions (emergency buttons, sensors, etc.), checking of possible loose parts and general adjustments are recommended.

This equipment has moving parts that collide with each other, creating risks of crushing. Extreme care should be taken when operating or handling the machine. Do not wear loose clothing, bracelets, necklaces, loose hair, or objects when operating or handling the machine, as there is a risk of getting caught in moving parts by pulling into the machine. This is automatic equipment that can perform movements during the process without human intervention. Always stay at a safe distance from moving mechanisms and avoid accidents.

This equipment requires a three-phase 208 VAC power line (as specified on the machine's nameplate) that can cause death. The electrical panel should only be accessed by authorized persons. You should not touch any electrical component with wet or sweaty hands. Remember that the static energy present in the human body can also damage some electro-electronic component, avoid touching them. Maintenance and repairs should only be carried out by qualified personnel or with appropriate training.

Looking forward to the complete safety and qualification of a new person in charge of the machine, Raumak offers courses designed to train operators in the correct operation and maintenance of the machine, and it is vital that the operator participate in and complete the completely safe training of the understanding of the entire procedure as well as the equipment as a whole.

If any instruction or procedure is not clear enough, contact Raumak prior to commencing operation on the machine.



What you should not do:

- Climb onto the conveyor (s) and/or on the machine;
- Disregard warning signs and safety signs;
- Bypass the safety system(s);
- Place limbs, hair and/or clothes between moving parts;
- Any service or maintenance on the equipment for which you have not been trained or are not qualified;
- Remove guards from the machine such as doors, safety relay, emergency buttons or interlock buttons.

The purpose of this is to alert operating and maintenance personnel to the possible risks found in the equipment.

Failure to observe this practice may result in serious injury or damage to the equipment not covered by the warranty.

This manual is written to be used by persons with adequate training or technical qualification to operate this type of equipment.



4.4 Safety Adhesives

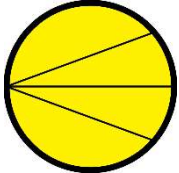
Visual signals indicate good practice and/or an actual hazardous situation that may involve damage to persons or property and require a human response to eliminate or control the hazard. Please familiarize yourself with the positions of these warnings and their meanings. If these warnings are damaged and/or illegible, they must be replaced.

SAFETY INSTRUCTIONS

- 1- Before starting the operation, please read warning and manuals fixed on the equipment.
- 2- This machine is driven by PLC. NEVER stay near of moving or rotating equipment parts.
- 3- Do not operate the machine before installation of all the safety devices.
- 4- For operation of this machine please do not wear loose clothes, rings/jewelry. Keep hair far from moving and rotating parts of the equipment.
- 5- Operating the equipment, please always use safety glasses and ear plugs.
- 6- Installation and Service/Technical Assistance of this equipment must be carried out by an authorized Raumak technician.
- 7- Before any cleaning procedure, repair or components replacement, please switch of all the power inputs and make sure that each one is shielded by locking device.
- 8- Proper operation of this machine is full responsibility of the operator, who must be following the procedures given in trainings and instructions of the manual as well as warning stickers on the machine.
- 9- If you have any doubt regarding operation, please inform your boss immediately or contact the Raumak Service through the telephone +55 47 3370-4540 or email service@raumak.com.br

PLEASE DO NOT REMOVE THIS NOTICE



Sign	Explication / Orientation
	Adhesive that indicates information for a better traction of the driving belts Location on the machine: Pulling Belt.
	Adhesive that indicates potential risk of crushing. Location of potential risk on the machine: Conveyors rolls.
	Adhesive that indicates potential risk area of burning during and after the machine's operation. Locations of potential risk on the machine: Horizontal Sealing Assembly (Seal Metals); Vertical Sealing Assembly (Seal Metal).
	Adhesive that indicates potential risk area of crushing during the machine's operation. Locations of potential risk on the machine: Superior Mechanism (Drop Doors); Horizontal Sealing Assembly (Jaw); Vertical Sealing Assembly (Vertical Seal); Pulling Belts(Drive Belt).
	Indication of the pneumatic panel. Extreme care should be taken with respect to the air supply and sudden movements in pneumatic components during operation. Maintenance should performed only by qualified technicians.
	Indication of potential risk of electric shock. This equipment works with high voltage. All maintenance and inspections involving electrical components must be carried out by qualified personnel.
	Indication topics described in this manual must be read with great care, since they are related to the correct operation of the equipment and may even cause damage to the machine / process, or the operator.

4.5 Safety Devices

In order to prevent accidents, the machines are equipped with safety devices, if you make any changes to this device, it is your responsibility to enable its correct operation again, before starting operation with the machine, it is recommended that these be checked.

4.5.1 Security keys

Entrance, side and rear door, if any, when open, the security key does not allow the operation of the equipment.

4.5.2 General key

The general key allows the electricity to be turned on or off, it is mentioned as a safety device, it is strongly recommended that it is in the OFF position before any repair / service of the equipment.

4.5.3 Emergency button

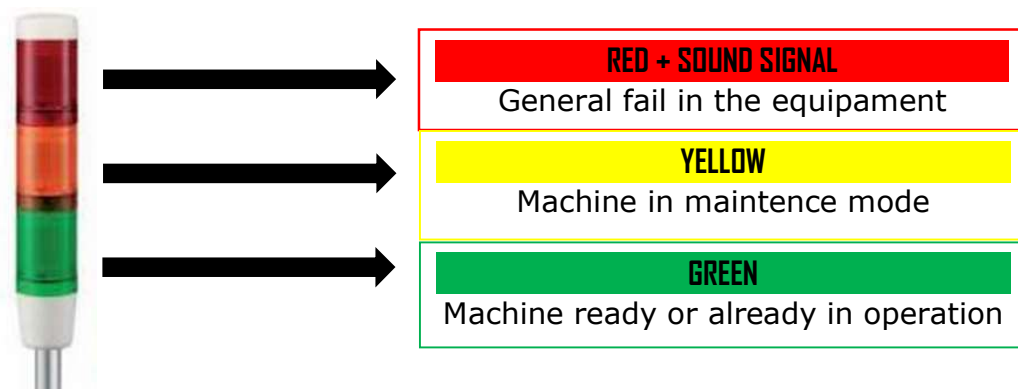
The emergency button is strategically located on the control panel (front) and pressing band for the complete closing of the machine in case of any adverse situation.

4.5.4 Automatic and manual depressurizer

The Automatic and Manual Depressurizer cuts off the pneumatic supply, eliminating risks related to the operation of the equipment's cylinders.

4.5.5 Light indication

The Luminous Indications are intended to help the operator in what condition the equipment is, it is extremely important that the risks and alerts in a machine are signaled in a visual and acoustic way in order to prevent accidents. The signalling column is able to perform this task, with an individual indication for each machine state. Below is the meaning of each colour.





5 CLEANING

5.1 Cleaning The Machine

To perform the cleaning procedure, the machine must be completely switched off (electric and pneumatic). The person responsible for the work must be completely sure that the equipment does not pose any risk. Daily cleaning of the equipment using a cloth with neutral products is recommended.

5.2 Cleaning Of Acrylics

The acrylic components cannot be cleaned with alcohol or products based on benzine, benzene, acetone or acids, as internal cracks may occur. When cleaning, only water, neutral soap or detergent should be used, avoiding the use of a sponge or rough cloth. Afterwards, polish with a soft flannel. For information purposes, you will find the adhesive below in acrylic pieces.

